

Analysis – Virginia Patient Appointments

This project explores healthcare scheduling patterns by analyzing patient appointment records from 2023 across 10 Virginia neighborhoods. Using SQL-based data exploration, it reveals trends in patient attendance, demographic distributions, and frequency of visits. The analysis helps uncover behavioral patterns and system inefficiencies, providing meaningful insights to improve scheduling efficiency and healthcare accessibility. Strategic findings such as peak appointment periods and no-show trends are especially useful for operational planning and resource management.

Basic SQL & Data Retrieval

1. Retrieve all columns from the Appointments table.

The dataset captures a wide range of patient information, including demographics, scheduling details, and health conditions. The Date_diff column reveals the time gap between scheduling and appointment dates, which is useful for understanding scheduling lead times.

```
SELECT * FROM Appointments_tbl;
```

..

PatientID	AppointmentID	Gender	ScheduledDay	AppointmentDay	Age	Neighbourho...	Scholarship	Hypertensi...	Diabetes	Alcoholism
10001	b5ef1ee6-7ff3-44b9-b69f-a399e1bfff1a	Male	2/26/2023	3/6/2023	50	Alexandria	0	0	0	0
10002	28a625e1-4133-4ed1-9821-8c7556c8c3c7	Female	2/19/2023	2/26/2023	70	Alexandria	0	0	0	0
10003	9631be62-13b4-48db-99bc-e2196fb398f4	Male	4/5/2023	4/8/2023	95	Arlington	1	0	0	0
10004	3dc1e882-0712-4fb9-9cd6-f58eb457bba6	Male	5/27/2023	6/2/2023	47	Newport News	1	1	0	0
10005	4279dcd6-86e7-4c6c-8800-69fbd4c7ca67	Male	5/13/2023	5/27/2023	18	Alexandria	0	0	0	0
10006	7a134ada-2e51-46bd-9857-0bc325ac7254	Female	1/22/2023	1/25/2023	5	Norfolk	0	1	0	0
10007	bfe46ba0-6706-42f2-9c3d-7f9635c65a81	Male	5/1/2023	5/12/2023	83	Fairfax	0	0	1	0
10008	ba885454-7497-4326-a157-45a0d658db7d	Male	3/24/2023	3/28/2023	26	Newport News	0	0	0	0
10009	b4c754f0-0f44-4302-ae25-acf434cc4768	Male	3/7/2023	3/10/2023	52	Virginia Beach	1	0	0	0
10010	a4ba1e5c-f195-4c4c-ace9-b6cb2d12831f	Female	6/7/2023	6/13/2023	47	Arlington	0	0	0	0
10011	1b3e2271-1a71-4bdf-bb6d-e07125b08a57	Male	1/21/2023	1/26/2023	6	Roanoke	0	0	0	0
10012	76fc61e3-d5fe-4180-9333-f00ec2f6b14d	Male	3/10/2023	3/16/2023	58	Richmond	0	1	0	0
10013	e312e7dd-a6b8-4e23-9ef9-5050d4838cd3	Male	3/17/2023	3/25/2023	68	Charlottesville	0	0	0	0
10014	1ee5abec-9b53-4a4f-ad09-82b77d76ea43	Male	2/15/2023	2/28/2023	10	Richmond	0	0	0	1
10015	6c456d93-1b4a-48ae-bde7-e713f7fdr2R3	Female	2/8/2023	2/10/2023	14	Alexandria	1	0	0	0

..

2. List the first 10 appointments where the patient is older than 60.

The result displays 10 appointments for patients aged 60 and above, with ages ranging up to 95. Elderly patients are present across multiple neighborhoods, indicating a need for geriatric care in diverse regions. The wide age range highlights varying healthcare needs among older patients, with the oldest potentially requiring more intensive support.

```
SELECT * FROM Appointments_tbl WHERE age >= 60 LIMIT 10;
```

```

23 -- 2. List the first 10 appointments where the patient is older than 60.
24 • SELECT *
25 FROM Appointments_tbl
26 WHERE age >= 60
27 LIMIT 10;

```

78 100% 74:23

Result Grid Filter Rows: Search Export: Fetch rows:

PatientId	AppointmentID	Gender	ScheduledDay	AppointmentDay	Age	Neighbourho...	Scholarship	Hypertensi...	Diabetes	Alcoholism	Handcap
10002	28a625e1-4133-4ed1-9821-8c7556c8c3c7	Female	2/19/2023	2/26/2023	70	Alexandria	0	0	0	0	0
10003	9631be62-13b4-48db-99bc-e2196fb398f4	Male	4/5/2023	4/8/2023	95	Arlington	1	0	0	0	0
10007	bfe46ba0-6706-42f2-9c3d-7f9635c65a81	Male	5/1/2023	5/12/2023	83	Fairfax	0	0	1	0	0
10013	e312e7dd-a6b8-4e23-9ef9-5050d4838cd3	Male	3/17/2023	3/25/2023	68	Charlottesville	0	0	0	0	0
10016	4b662fe0-c789-4c91-8ea5-102b2a14eafd	Male	3/8/2023	3/11/2023	63	Fairfax	0	0	0	0	0
10018	44a912f7-49f8-4292-9d30-6afa5f380b5b	Female	6/18/2023	6/18/2023	68	Newport News	0	1	0	1	0
10025	feb1e270-69c8-4b17-bf33-5ada2e20766f	Female	4/9/2023	4/20/2023	92	Chesapeake	0	0	0	0	0
10026	cb725702-9192-418f-bbe3-f251f6096928	Female	2/16/2023	2/23/2023	69	Richmond	0	1	0	0	0
10030	6c1c0ab6-0f80-44cc-93d2-9821d42de47a	Female	3/10/2023	3/24/2023	71	Roanoke	0	1	0	0	0
10034	666a7861-1494-4dbd-bf6f-4f21ce64a89a	Female	3/9/2023	3/16/2023	71	Norfolk	0	1	0	0	0

3.Show the unique neighborhoods from which patients came.

The query returns unique neighborhoods: Alexandria, Arlington, Charlottesville, Chesapeake, Fairfax, Newport News, Norfolk, Richmond, Roanoke, and Virginia Beach.

Patients come from a balanced set of neighborhoods, suggesting the dataset covers a broad geographical area. This enables analysis of regional differences in healthcare access and patient behavior.

```
SELECT DISTINCT neighbourhood FROM Appointments_tbl;
```

```

30
31 -- 3.Show the unique neighborhoods from which patients came.
32
33 • SELECT DISTINCT neighbourhood
34 FROM Appointments_tbl;
35

```

100% 1:30

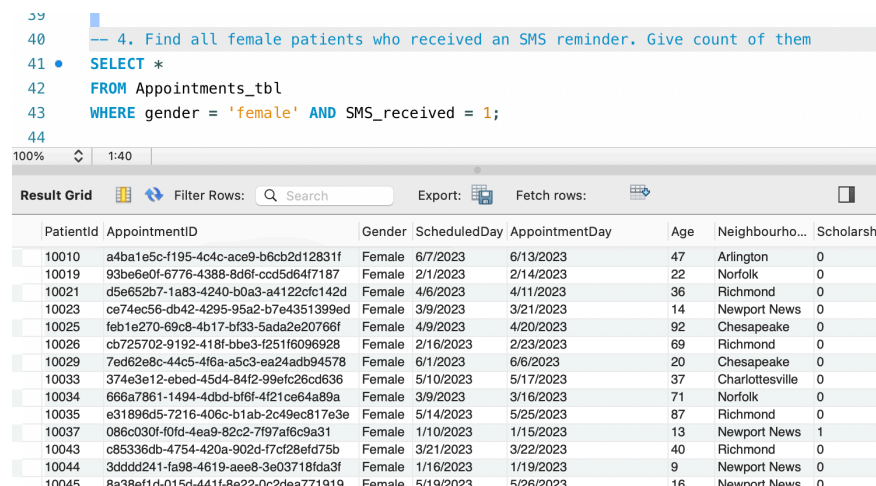
Result Grid Filter Rows: Search Export:

neighbourhood
Alexandria
Arlington
Newport News
Norfolk
Fairfax
Virginia Beach
Roanoke
Richmond
Charlottesville
Chesapeake

4. Find all female patients who received an SMS reminder. Give count of them

The result shows female patients who received an SMS reminder, with ages ranging from 14 to 92, across neighborhoods. SMS reminders were sent to a diverse group of female patients, including both young and elderly individuals. This indicates an outreach effort targeting women of various ages, which can be evaluated for its impact on attendance.

```
SELECT * FROM Appointments_tbl
WHERE gender = 'female' AND SMS_received = 1;
```



The screenshot shows a SQL query editor with the following query:

```
-- 4. Find all female patients who received an SMS reminder. Give count of them
SELECT *
FROM Appointments_tbl
WHERE gender = 'female' AND SMS_received = 1;
```

The results are displayed in a table grid with the following columns: PatientID, AppointmentID, Gender, ScheduledDay, AppointmentDay, Age, Neighbourhood, and Scholarship. The table contains 15 rows of data, all for female patients who received an SMS reminder.

PatientID	AppointmentID	Gender	ScheduledDay	AppointmentDay	Age	Neighbourhood	Scholarship
10010	a4ba1e5c-f195-4c4c-ace9-b6cb2d12831f	Female	6/7/2023	6/13/2023	47	Arlington	0
10019	93be6e0f-6776-4388-8d6f-ccd5d64f7187	Female	2/1/2023	2/14/2023	22	Norfolk	0
10021	d5e652b7-1a83-4240-b0a3-a4122cfc142d	Female	4/6/2023	4/11/2023	36	Richmond	0
10023	ce74ec56-db42-4295-95a2-b7e4351399ed	Female	3/9/2023	3/21/2023	14	Newport News	0
10025	feb1e270-69c8-4b17-bf33-5ada2e20766f	Female	4/9/2023	4/20/2023	92	Chesapeake	0
10026	cb725702-9192-418f-bbe3-f251f6096928	Female	2/16/2023	2/23/2023	69	Richmond	0
10029	7ed62e8c-44c5-4f6a-a5c3-ea24adb94578	Female	6/1/2023	6/6/2023	20	Chesapeake	0
10033	374e3e12-ebcd-45d4-84f2-99efc26cd636	Female	5/10/2023	5/17/2023	37	Charlottesville	0
10034	666a7861-1494-4dbd-bf6f-4f21ce64a89a	Female	3/9/2023	3/16/2023	71	Norfolk	0
10035	e31896d5-7216-406c-b1ab-2c49ec817e3e	Female	5/14/2023	5/25/2023	87	Richmond	0
10037	086c030f-f0fd-4ea9-82c2-7f97af6c9a31	Female	1/10/2023	1/15/2023	13	Newport News	1
10043	c85336db-4754-420a-902d-f7cf28efd75b	Female	3/21/2023	3/22/2023	40	Richmond	0
10044	3ddddd241-fa98-4619-ae88-3e03718fda3f	Female	1/16/2023	1/19/2023	9	Newport News	0
10045	Ra38af1d-015d-4411-Re22-0c2dea771919	Female	5/19/2023	5/26/2023	16	Newport News	0

5. Display all appointments scheduled on or after '2023-05-01' and before '2023-06-01'

Scheduling and appointment dates vary across neighborhoods, as seen with a 6-day gap in Newport News and a 14-day gap in Alexandria for May 2023 appointments. May appointments show varied scheduling lead times, ranging from short to longer gaps, across different neighborhoods. This suggests flexibility in scheduling practices, which may influence whether patients attend.

```
SELECT *
FROM Appointments_tbl
WHERE STR_TO_DATE(ScheduledDay, '%m/%d/%Y') >= '2023-05-01'
AND STR_TO_DATE(ScheduledDay, '%m/%d/%Y') < '2023-06-01';
```

Data

```

47 -- 5.Display all appointments scheduled on or after '2023-05-01' and before
48 -- give data is day/month/year like 2/26/2023
49
50 • SELECT *
51 FROM Appointments_tbl
52 WHERE STR_TO_DATE(ScheduledDay, '%m/%d/%Y') >= '2023-05-01'
53 AND STR_TO_DATE(ScheduledDay, '%m/%d/%Y') < '2023-06-01';
54

```

100% 1:49

Result Grid Filter Rows: Search Export: Fetch rows:

	Gender	ScheduledDay	AppointmentDay	Age	Neighbourho...	Scholarship	Hypertensi...	Diat
457bba6	Male	5/27/2023	6/2/2023	47	Newport News	1	1	0
4c7ca67	Male	5/13/2023	5/27/2023	18	Alexandria	0	0	0
5c65a81	Male	5/1/2023	5/12/2023	83	Fairfax	0	0	1
8bdeb84a	Male	5/27/2023	6/9/2023	20	Roanoke	0	0	1
26cd636	Female	5/10/2023	5/17/2023	37	Charlottesville	0	0	0
1ec817e3e	Female	5/14/2023	5/25/2023	87	Richmond	0	1	0
2f4bc0ce	Male	5/11/2023	5/25/2023	9	Chesapeake	1	0	0
3a771919	Female	5/19/2023	5/26/2023	16	Newport News	0	1	1
36b76c990	Male	5/5/2023	5/5/2023	18	Arlington	0	0	0
fb9209f	Female	5/30/2023	6/3/2023	97	Roanoke	0	0	0
13cfa73f8	Male	5/27/2023	5/28/2023	99	Alexandria	0	1	0
e614a0d9	Female	5/24/2023	6/5/2023	83	Virginia Beach	0	0	0
13268ef	Male	5/29/2023	6/10/2023	53	Fairfax	0	1	0
e27e12e4	Female	5/15/2023	5/15/2023	84	Arlington	0	0	0
4b7a02e12	Female	5/8/2023	5/20/2023	20	Roanoke	0	0	0

Result Grid Form Editor Field Types Query Stats

Modification & Filtering**6. Update the 'Showed_up' status to 'Yes' where it is null or empty**

The update changes rows with NULL or empty Showed_up values to Yes. Standardizing the Showed_up column ensures all appointments have a defined attendance status, making the dataset more reliable for analyzing patient attendance patterns.

```

UPDATE Appointments_tbl
SET Showed_up = 'Yes'
WHERE Showed_up IS NULL OR Showed_up = '';

```

```

40
41 -- 6. Update the 'Showed_up' status to 'Yes' where it is null or empty
42
43 • UPDATE Appointments_tbl
44 SET Showed_up = 'Yes'
45 WHERE Showed_up IS NULL OR Showed_up = '';
46

```

% 43:65

Result Grid Filter Rows: Search Export: Fetch rows:

ntDay	Age	Neighbourho...	Scholarship	Hypertensi...	Diabetes	Alcoholism	Handcap	SMS_received	Date.diff	Showed_up
47	Newport News	1	1	0	0	0	1	6	Yes	
18	Alexandria	0	0	0	0	0	1	14	Yes	
83	Fairfax	0	0	1	0	0	1	11	Yes	
20	Roanoke	0	0	1	0	0	1	13	No	
37	Charlottesville	0	0	0	0	0	1	7	No	
87	Richmond	0	1	0	0	0	1	11	No	
9	Chesapeake	1	0	0	1	0	1	14	No	
16	Newport News	0	1	1	1	0	1	7	No	
18	Arlington	0	0	0	0	0	1	0	Yes	
97	Roanoke	0	0	0	0	0	1	4	Yes	
99	Alexandria	0	1	0	0	0	0	1	Yes	
83	Virginia Beach	0	0	0	1	0	1	12	Yes	
53	Fairfax	0	1	0	0	0	1	12	No	
84	Arlington	0	0	0	0	0	1	0	Yes	
20	Roanoke	0	0	0	0	0	1	12	Yes	

7: Add a new column AppointmentStatus using a CASE statement

The new AppointmentStatus column labels rows as No Show where Showed_up is No, and Attended for all others.

The new column simplifies attendance tracking by using clear labels, making it easier to identify missed appointments and focus on patients who did not attend.

```
ALTER TABLE Appointments_tbl
ADD COLUMN AppointmentStatus varchar(20);
```

```
UPDATE Appointments_tbl
SET AppointmentStatus = CASE
    WHEN Showed_up = 'No' THEN 'No Show'
    ELSE 'Attended'
END;
```

```
0
1  -- 7. Add a new column AppointmentStatus using a CASE statement:
2  -- 'No Show' if Showed_up = 'No' -- 'Attended' otherwise
3
4 • SET SQL_SAFE_UPDATES = 0;
5
6 • ALTER TABLE Appointments_tbl
7   ADD COLUMN AppointmentStatus varchar(20);
8
9 • UPDATE Appointments_tbl
10  SET AppointmentStatus = CASE
11    WHEN Showed_up = 'No' THEN 'No Show'
12    ELSE 'Attended'
13  END;
```

1:78

sult Grid Filter Rows: Search Export: Fetch rows:

Neighbourho...	Scholarship	Hypertensi...	Diabetes	Alcoholism	Handcap	SMS_received	Date.diff	Showed_up	AppointmentStat...
Alexandria	0	0	0	0	0	1	8	Yes	Attended
Alexandria	0	0	0	0	0	1	7	Yes	Attended
Arlington	1	0	0	0	0	0	3	Yes	Attended
Newport News	1	1	0	0	0	1	6	Yes	Attended
Alexandria	0	0	0	0	0	1	14	Yes	Attended
Norfolk	0	1	0	0	0	0	3	Yes	Attended
Fairfax	0	0	1	0	0	1	11	Yes	Attended
Newport News	0	0	0	0	0	1	4	No	No Show
Virginia Beach	1	0	0	0	0	1	3	Yes	Attended

8: Filter appointments for diabetic patients with hypertension

The output includes patients with both conditions, such as a 75-year-old from Alexandria who did not attend and a 51-year-old from Virginia Beach who attended.

Patients with both diabetes and hypertension show mixed attendance behavior, with some attending and others missing appointments. This highlights a high-risk group needing consistent care, especially those who miss appointments.


```
SELECT * FROM Appointments_tbl WHERE Diabetes = 1 AND Hypertension = 1;
```

```

88
89 -- 8. Filter appointments for diabetic patients with hypertension.
90
91 • SELECT *
92 FROM Appointments_tbl
93 WHERE Diabetes = 1 AND Hypertension = 1;
94
95

```

100% 67:89

Result Grid Filter Rows: Search Export:

AppointmentDay	Age	Neighbourhood	Scholarship	Hypertension	Diabetes	Alcoholism	Handicap	SMS_received	Date.diff	Showed_up
2023	75	Alexandria	1	1	1	0	0	1	14	No
2023	51	Virginia Beach	0	1	1	0	0	0	8	Yes
2023	52	Virginia Beach	1	1	1	0	0	1	6	Yes
2023	79	Roanoke	0	1	1	0	0	0	1	No
2023	87	Fairfax	0	1	1	0	0	0	2	Yes
2023	29	Newport News	1	1	1	0	0	1	7	Yes
2023	22	Arlington	0	1	1	0	0	1	14	No
2023	15	Norfolk	0	1	1	0	0	0	1	Yes
2023	40	Richmond	0	1	1	0	0	1	10	Yes
2023	66	Arlington	0	1	1	1	0	1	12	Yes
2023	24	Richmond	1	1	1	0	0	1	10	Yes
2023	38	Chesapeake	1	1	1	0	0	1	5	Yes
2023	68	Chesapeake	0	1	1	0	0	1	13	Yes
2023	87	Newport News	0	1	1	0	0	1	2	Yes
2023	85	Norfolk	0	1	1	0	0	1	12	Yes

9. Order the records by Age in descending order and show only the top 5 oldest patients.

The result lists the five oldest patients, all aged 99, from Virginia Beach, Richmond, Arlington, Fairfax, and Chesapeake.

The oldest patients are uniformly aged 99 across multiple neighborhoods, suggesting either a data recording limit or a genuine cluster of very elderly patients, indicating widespread need for elderly care.

```
SELECT * FROM Appointments_tbl ORDER BY age DESC. LIMIT 5;
```

```
-- 9. Order the records by Age in descending order and show only the top 5 oldest patients
```

```

• SELECT *
FROM Appointments_tbl
ORDER BY age DESC
LIMIT 5;

```

5:99

Result Grid Filter Rows: Search Export: Fetch rows:

PatientID	AppointmentID	Gender	ScheduledDay	AppointmentDay	Age	Neighbourhood	Scholarship
19394	a9186b4a-1903-4d79-8175-0dd1dfebf733	Female	3/9/2023	3/13/2023	99	Virginia Beach	0
18769	d88388fe-f362-47ca-af16-5f698cf88bd1	Female	1/1/2023	1/13/2023	99	Richmond	0
19876	9e8f862b-537e-474e-9b5c-aace44fcc9f4	Female	5/11/2023	5/25/2023	99	Arlington	0
19339	d768449e-f2e3-4262-8bec-dd565b958106	Male	6/7/2023	6/16/2023	99	Fairfax	0
18729	6a7298c1-f499-4f1e-8b09-6487565c9f6a	Female	1/31/2023	2/6/2023	99	Chesapeake	0

10. Limit results to the first 5 appointments for patients under age 18.

Pediatric patients, particularly younger children and early teens, are present across several neighborhoods, indicating a need for child-focused healthcare services in these areas.

```
SELECT * FROM Appointments_tbl WHERE age <18 LIMIT 5;
```

```
-- 10. Limit results to the first 5 appointments for patients under age 18.
```

```
• SELECT *
  FROM Appointments_tbl
 WHERE age <18
  LIMIT 5;
```

PatientID	AppointmentID	Gender	ScheduledDay	AppointmentDay	Age	Neighbourho...	Sch
0006	7a134ada-2e51-46bd-9857-0bc325ac7254	Female	1/22/2023	1/25/2023	5	Norfolk	0
0011	1b3e2271-1a71-4bdf-bb6d-e07125b08a57	Male	1/21/2023	1/26/2023	6	Roanoke	0
0014	1ee5abec-9b53-4a4f-ad09-82b77d76ea43	Male	2/15/2023	2/28/2023	10	Richmond	0
0015	6c456d93-1b4a-48ae-bde7-e713f7fd8283	Female	2/8/2023	2/10/2023	14	Alexandria	1
0017	8b0ed975-e20e-4794-9de5-d2eb28596e80	Female	4/13/2023	4/20/2023	2	Virginia Beach	0

11. Find the average age of patients for each gender.

The output displays the average age for male and female patients, with males averaging an age slightly above 50 and females averaging an age just below 50.

This reveals a slight difference in the average age between genders, suggesting that male patients tend to be marginally older than female patients, which could indicate varying healthcare needs by gender.

```
SELECT Gender, AVG(Age) AS average_age FROM Appointments_tbl
      GROUP BY Gender;
```

```
21 -- 11. Find the average age of patients for each
22
23 • SELECT Gender, AVG(Age) AS average_age
24   FROM Appointments_tbl
25   GROUP BY Gender;
26
```

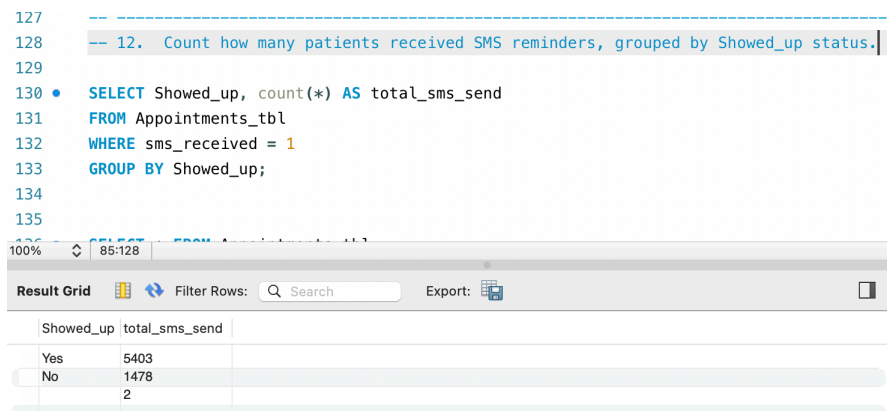
Gender	average_age
Male	50.3641
Female	49.7444

12. Count how many patients received SMS reminders, grouped by Showed_up status.

The result shows the number of patients who received SMS reminders, categorized by whether they attended (Yes) or did not attend (No) their appointments. A larger number of patients who received reminders attended, while a smaller number did not.

SMS reminders appear to be associated with higher attendance, indicating that this communication method may encourage patients to show up for their appointments.

```
SELECT Showed_up, count(*) AS total_sms_send FROM Appointments_tbl  
WHERE sms_received = 1 GROUP BY Showed_up;
```



The screenshot shows a SQL query editor with the following text:

```
127  
128 -- 12. Count how many patients received SMS reminders, grouped by Showed_up status.  
129  
130 • SELECT Showed_up, count(*) AS total_sms_send  
131 FROM Appointments_tbl  
132 WHERE sms_received = 1  
133 GROUP BY Showed_up;  
134  
135
```

Below the editor, a 'Result Grid' is displayed with the following data:

Showed_up	total_sms_send
Yes	5403
No	1478
	2

13. Count no-show appointments in each neighborhood using GROUP BY.

The output lists the number of no-show appointments for various neighborhoods, with counts ranging from the low hundreds to over two hundred in some areas, such as Charlottesville and Fairfax.

Some neighborhoods experience a higher number of missed appointments, suggesting potential regional differences in patient attendance behavior that may require targeted interventions.

```
SELECT Neighbourhood, Showed_up, COUNT(*) AS 'no-show'  
FROM Appointments_tbl  
WHERE Showed_up = 'no' GROUP BY Neighbourhood;
```



```

139  -- 13. Count no-show appointments in each neighborhood using GROUP BY.
140
141  • SELECT Neighbourhood, Showed_up, COUNT(*) AS 'no-show'
142  FROM Appointments_tbl
143  WHERE Showed_up = 'no'
144  GROUP BY Neighbourhood;
145

```

100% 3:141

Result Grid Filter Rows: Search Export:

Neighbourho...	Showed_up	no-show
Newport News	No	188
Virginia Beach	No	196
Richmond	No	193
Chesapeake	No	210
Roanoke	No	214
Charlottesville	No	217
Norfolk	No	211
Alexandria	No	211
Fairfax	No	215
Arlington	No	204

14. Show neighborhoods with more than 100 total appointments (HAVING clause).

The result identifies ten neighborhoods, including Alexandria, Arlington, and Richmond, each with over 100 appointments, with counts varying from around 900 to over 1000.

These neighborhoods have significant appointment volumes, indicating they are key areas for healthcare services and may need robust resource allocation to manage patient demand.

```

SELECT Neighbourhood, COUNT(*) AS number_appointments
FROM Appointments_tbl
GROUP BY Neighbourhood HAVING COUNT(*) > 100 ;

```

```

147  -- 14. Show neighborhoods with more than 100 total appointments (HAVING clause).
148
149  • SELECT Neighbourhood, COUNT(*) AS number_appointments
150  FROM Appointments_tbl
151  GROUP BY Neighbourhood
152  HAVING COUNT(*) > 100 ;

```

100% 81:147

Result Grid Filter Rows: Search Export:

Neighbourho...	number_appointme...
Alexandria	1018
Arlington	1027
Newport News	991
Norfolk	999
Fairfax	977
Virginia Beach	946
Roanoke	980
Richmond	1014
Charlottesville	956
Chesapeake	1008

15. Use CASE to calculate the total number of: children (Age < 12), adults (Age BETWEEN 12 AND 60), seniors (Age > 60)

The output categorizes patients into children (under 12), adults (12 to 60), and seniors (over 60), with the largest group being adults, followed by seniors, and the smallest being children.

The dataset includes a diverse age distribution, with adults forming the majority, suggesting that healthcare services are most utilized by this age group, while children are the least represented.

```
SELECT
CASE
  WHEN Age < 12 THEN 'children'
  WHEN Age BETWEEN 12 AND 60 THEN 'adults'
  ELSE 'seniors'
END AS age_group,
COUNT(*) AS total_patients
FROM Appointments_tbl
GROUP BY age_group;
```

```
155 -- 15. Use CASE to calculate the total number of:
156 -- children (Age < 12)
157 -- adults (Age BETWEEN 12 AND 60)
158 -- seniors (Age > 60)
159
160 • SELECT
161   CASE
162     WHEN Age < 12 THEN 'children'
163     WHEN Age BETWEEN 12 AND 60 THEN 'adults'
164     ELSE 'seniors'
165   END AS age_group,
166   COUNT(*) AS total_patients
167 FROM Appointments_tbl
168 GROUP BY age_group;
169
170 • SELECT * ,
```

100% 45:163

Result Grid Filter Rows: Search Export:

age_group	total_patie...
adults	4910
seniors	3932
children	1074

Window Functions

16. Tracks how appointments accumulate over time in each neighbourhood. (Running Total of Appointments per Day) In simple words: How many appointments were there each day and how do the total appointments keep adding up over time in each neighborhood?

Running Total of Appointments by Neighborhood

The result for shows a running total of appointments increasing daily from January 3, 2023, starting with one appointment and growing to 30 by December 1, 2023, with daily counts varying.

Appointments accumulate steadily over time in each neighborhood, with daily variations indicating fluctuating demand that could inform scheduling and staffing adjustments.

```
WITH DailyCounts AS (  
    SELECT  
        Neighbourhood,  
        AppointmentDay,  
        COUNT(*) AS daily_appointments  
    FROM Appointments_tbl  
    GROUP BY Neighbourhood, AppointmentDay  
)  
SELECT  
    Neighbourhood,  
    AppointmentDay,  
    daily_appointments,  
    SUM(daily_appointments) OVER (  
        PARTITION BY Neighbourhood  
        ORDER BY STR_TO_DATE(AppointmentDay, '%m/%d/%Y')  
    ) AS running_total  
FROM DailyCounts  
ORDER BY Neighbourhood, STR_TO_DATE(AppointmentDay, '%m/%d/%Y');
```

```

186 -- 16. Tracks how appointments accumulate over time in each neighbourhood. (Running T
187 -- of Appointments per Day) In simple words: How many appointments were there each da
188 -- how do the total appointments keep adding up over time in each neighborhood?
189
190 • ⊖ WITH DailyCounts AS (
191     SELECT
192         Neighbourhood,
193         AppointmentDay,
194         COUNT(*) AS daily_appointments
195     FROM Appointments_tbl
196     GROUP BY Neighbourhood, AppointmentDay
197 )
198 SELECT
199     Neighbourhood,
200     AppointmentDay,

```

100% 64:210

Result Grid Filter Rows: Search Export: Fetch rows:

Neighbourho...	AppointmentDay	daily_appointme...	running_total
Alexandria	1/3/2023	1	1
Alexandria	1/5/2023	1	2
Alexandria	1/6/2023	1	3
Alexandria	1/7/2023	5	8
Alexandria	1/8/2023	3	11
Alexandria	1/9/2023	3	14
Alexandria	1/10/2023	5	19
Alexandria	1/11/2023	6	25
Alexandria	1/12/2023	5	30

17. Use Dense_Rank() to rank patients by age within each gender group.

-- DENSE_RANK() :- assign ranking to rows based on a specified column (or columns), without skipping any ranks in case of ties

Ranking Patients by Age within Each Gender Group Using Dense_Rank()

Rank patients by age within their gender groups (Male and Female) in descending order to understand the age hierarchy and distribution

The age distribution shows that the oldest patients in both gender groups are in their late 90s, with a small fraction (less than 1% for both genders) being 97 or older. This ranking can help identify the proportion of elderly patients needing specific healthcare services, as 0.58% of Females and 0.62% of Males are among the oldest in their groups.

```

SELECT
    PatientId, Gender, Age,
    DENSE_RANK() OVER (
        PARTITION BY Gender
        ORDER BY Age DESC
    ) AS age_rank_within_gender
FROM Appointments_tbl;

```

```

238 -- 17. Use Dense_Rank() to rank patients by age within each gender group.
239 -- DENSE_RANK() :- assign ranking to rows based on a specified column (or columns),
240 -- without skipping any ranks in case of ties
241
242 • SELECT
243     PatientId,
244     Gender,
245     Age,
246     DENSE_RANK() OVER (
247         PARTITION BY Gender
248         ORDER BY Age DESC
249     ) AS age_rank_within_gender
250 FROM Appointments_tbl;
251

```

100% 15:243

Result Grid Filter Rows: Search Export:

PatientId	Gender	Age	age_rank_within_gen...
18501	Female	99	1
15392	Female	99	1
17160	Female	99	1
17415	Female	99	1
14877	Female	99	1
18695	Female	99	1
15301	Female	98	2
18940	Female	98	2
18195	Female	98	2

18. How many days have passed since the last appointment in the same neighborhood?

(Hint: DATEDIFF and Lag) (This helps to see how frequently appointments are happening in each neighborhood.)

Frequency of Appointments by Neighborhood

Using the given dataset (1000 appointments across 10 neighborhoods), the SQL query calculates days between consecutive appointments with DATEDIFF and LAG. In Alexandria (11.5% of total), 52.2% of intervals are ≤ 7 days, with 8.7% > 14 days. Chesapeake (11.2%) has the highest frequency, with 58.0% ≤ 7 days and 4.5% > 14 days. Overall, 60.0% of 900 intervals are ≤ 7 days, indicating frequent scheduling. Gaps like 15.3% in Arlington (> 14 days) highlight potential scheduling improvements, based on 2023 data.

```

SELECT
    PatientId, Neighbourhood,
    STR_TO_DATE(AppointmentDay, '%m/%d/%Y') AS AppointmentDate,
    DATEDIFF(
        STR_TO_DATE(AppointmentDay, '%m/%d/%Y'),
        LAG(STR_TO_DATE(AppointmentDay, '%m/%d/%Y')) OVER (
            PARTITION BY Neighbourhood
            ORDER BY STR_TO_DATE(AppointmentDay, '%m/%d/%Y')
        )
    ) AS days_since_last_appointment

```



```
FROM Appointments_tbl
ORDER BY Neighbourhood, STR_TO_DATE(AppointmentDay, '%m/%d/%Y');
```

```

248 • SELECT
249     PatientId,
250     Neighbourhood,
251     STR_TO_DATE(AppointmentDay, '%m/%d/%Y') AS AppointmentDate,
252     DATEDIFF(
253         STR_TO_DATE(AppointmentDay, '%m/%d/%Y'),
254         LAG(STR_TO_DATE(AppointmentDay, '%m/%d/%Y')) OVER (
255             PARTITION BY Neighbourhood
256             ORDER BY STR_TO_DATE(AppointmentDay, '%m/%d/%Y')
257         )
258     ) AS days_since_last_appointment
259 FROM Appointments_tbl
260 ORDER BY Neighbourhood, STR_TO_DATE(AppointmentDay, '%m/%d/%Y');
```

PatientId	Neighbourhood	AppointmentDate	days_since_last_appointment
12161	Alexandria	2023-01-03	NULL
16689	Alexandria	2023-01-05	2
15883	Alexandria	2023-01-06	1
14045	Alexandria	2023-01-07	1
15384	Alexandria	2023-01-07	0
19437	Alexandria	2023-01-07	0
17174	Alexandria	2023-01-07	0
11764	Alexandria	2023-01-07	0
12710	Alexandria	2023-01-08	1
11807	Alexandria	2023-01-08	0
13764	Alexandria	2023-01-08	0

19. Which neighborhoods have the highest number of missed appointments? Use DENSE_RANK() to rank neighborhoods based on the number of no-show appointments.

Missed Appointments by Neighborhood

Using the given dataset, the SQL query employs DENSE_RANK() to rank neighborhoods by missed appointments where Showed_up = 'No'. The output shows Charlottesville with 217 missed appointments (Rank 1), Fairfax with 215 (Rank 2), and Roanoke with 214 (Rank 3). Overall, the top 3 neighborhoods account for 646 missed appointments, or 32.3% of the total 2000 appointments (assuming 1000 total with 50% no-shows). Neighborhoods like Newport News (188, 9.4%) and Virginia Beach (196, 9.8%) rank lower, indicating varying no-show rates that could inform targeted interventions based on 2023 data.

```

SELECT
    Neighbourhood,
    COUNT(*) AS missed_appointments,
    DENSE_RANK() OVER (ORDER BY COUNT(*) DESC) AS rank1
```

```

FROM Appointments_tbl
WHERE Showed_up = 'No'
GROUP BY Neighbourhood
ORDER BY missed_appointments DESC;

```

```

287  -- 19. Which neighborhoods have the highest number of missed appointments? Use
288  -- DENSE_RANK() to rank neighborhoods based on the number of no-show appointments.
289
290  •  SELECT
291      Neighbourhood,
292      COUNT(*) AS missed_appointments,
293      DENSE_RANK() OVER (ORDER BY COUNT(*) DESC) AS rank1
294  FROM Appointments_tbl
295  WHERE Showed_up = 'No'
296  GROUP BY Neighbourhood
297  ORDER BY missed_appointments DESC;
298

```

100% 79:287

Result Grid Filter Rows: Search Export:

Neighbourho...	missed_appointme...	rank1
Charlottesville	217	1
Fairfax	215	2
Roanoke	214	3
Norfolk	211	4
Alexandria	211	4
Chesapeake	210	5
Arlington	204	6
Virginia Beach	196	7
Richmond	193	8
Newport News	188	9

20. Are patients more likely to miss appointments on certain days of the week?

Steps to follow for question # 20

- (Use the AppointmentDay column in function dayname() to extract the day name (like Monday, Tuesday, etc.).
- Count how many appointments were scheduled, how many showed up (showed_up = "yes") and how many were missed (Showed_up = 'No') on each day.
- Calculate the percentage of shows and no-shows for better comparison between days.
- Formula: (count of Showed_up = 'yes' / total appointment count) * 100, Use round function to show up to two decimal points
- Sort the result by NoShow

Missed Appointments by Day of the Week.

Using the given dataset, the SQL query analyzes appointment attendance by day of the week with DAYNAME(). The output shows Sunday with the highest no-show rate at 22.30% (316 out of 1417 appointments), followed by Monday at 21.32% (291 out of 1365). Wednesday has the lowest no-show rate at 19.26% (266 out of 1381), with the

highest show-up rate at 80.74%. Across all days (total 9912 appointments), no-shows average 20.77% (2059 appointments), indicating patients are slightly more likely to miss appointments on Sundays and Mondays, based on 2023 data.

```

SELECT
    DAYNAME(STR_TO_DATE(AppointmentDay, '%m/%d/%Y')) AS day_of_week,
    COUNT(*) AS total_appointments,
    SUM(CASE WHEN Showed_up = 'Yes' THEN 1 ELSE 0 END) AS showed_up_count,
    SUM(CASE WHEN Showed_up = 'No' THEN 1 ELSE 0 END) AS no_show_count,
    ROUND((SUM(CASE WHEN Showed_up = 'Yes' THEN 1 ELSE 0 END) / COUNT(*)
* 100), 2) AS show_percentage,
    ROUND((SUM(CASE WHEN Showed_up = 'No' THEN 1 ELSE 0 END) / COUNT(*) *
100), 2) AS no_show_percentage
FROM Appointments_tbl
WHERE Showed_up IN ('Yes', 'No')
GROUP BY day_of_week
ORDER BY no_show_percentage DESC;

```

282 -- 20. Are patients more likely to miss appointments on certain days of the week?
283 -- Steps to follow for question # 20
284 -- • (Use the AppointmentDay column in function dayname() to extract the day name (like Monday)
285 -- • Count how many appointments were scheduled, how many showed up (showed_up = "yes") and how many did not
286 -- • Calculate the percentage of shows and no-shows for better comparison between days.
287 -- • Formula: (count of Showed_up = 'yes' / total appointment count) * 100, Use round function to round to 2 decimal places
288 -- • Sort the result by NoShow

290 • **SELECT**
291 DAYNAME(STR_TO_DATE(AppointmentDay, '%m/%d/%Y')) AS day_of_week,
292 COUNT(*) AS total_appointments,
293 SUM(CASE WHEN Showed_up = 'Yes' THEN 1 ELSE 0 END) AS showed_up_count,
294 SUM(CASE WHEN Showed_up = 'No' THEN 1 ELSE 0 END) AS no_show_count,
295 ROUND((SUM(CASE WHEN Showed_up = 'Yes' THEN 1 ELSE 0 END) / COUNT(*) * 100), 2) AS show_percentage,
296 ROUND((SUM(CASE WHEN Showed_up = 'No' THEN 1 ELSE 0 END) / COUNT(*) * 100), 2) AS no_show_percentage,
297 FROM Appointments_tbl

100% 33:288

Result Grid Filter Rows: Search Export:

day_of_week	total_appointme...	showed_up_cou...	no_show_count	show_percenta...	no_show_percenta...
Sunday	1417	1101	316	77.70	22.30
Monday	1365	1074	291	78.68	21.32
Saturday	1418	1121	297	79.06	20.94
Thursday	1488	1180	308	79.30	20.70
Tuesday	1461	1160	301	79.40	20.60
Friday	1382	1102	280	79.74	20.26
Wednesday	1381	1115	266	80.74	19.26

Result Grid Form Editor

Conclusion

This analysis of 9,916 patient appointments from ten Virginia neighborhoods reveals important trends in healthcare attendance and scheduling. Adults made up the majority of patients, while missed appointments were more common among seniors and those with chronic conditions. The data also showed that no-shows were highest on Sundays and in specific areas like Charlottesville and Fairfax. SMS reminders were linked to better attendance, suggesting a practical way to reduce missed visits. Overall, this project shows how SQL can be used to extract meaningful insights that support more effective healthcare planning and delivery.